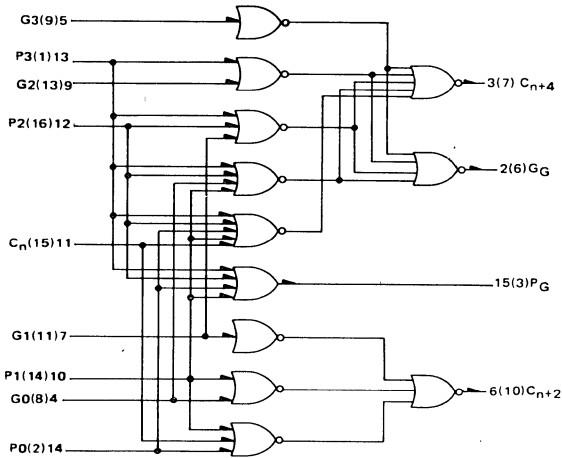


MC10179/MC10579

LOOK-AHEAD CARRY BLOCK



$P_D = 300 \text{ mW typ/pkg (No Load)}$ $V_{CC1} = \text{Pin 1 (5)}$
 $t_{pd} = 3.0 \text{ ns typ (Carry, Propagate)}$ $V_{CC2} = \text{Pin 16 (4)}$
 $= 4.0 \text{ ns typ (Generate)}$ $V_{EE} = \text{Pin 8 (12)}$

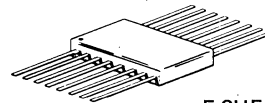
$P_G = P_0 + P_1 + P_2 + P_3$
 $G_G = (G_0 + P_1 + P_2 + P_3) (G_1 + P_2 + P_3) (G_2 + P_3) G_3$
 $C_{n+2} = (C_n + P_0 + P_1) (G_0 + P_1) G_1$
 $C_{n+4} = (C_n + P_0 + P_1 + P_2 + P_3) (G_0 + P_1 + P_2 + P_3) (G_1 + P_2 + P_3) (G_2 + P_3) G_3$



P SUFFIX
PLASTIC PACKAGE
CASE 648
MC10179 only



L SUFFIX
CERAMIC PACKAGE
CASE 620



F SUFFIX
CERAMIC PACKAGE
CASE 650
MC10579 only

The MC10179/MC10579 is a high speed, low power, standard MECL complex function that is designed to perform the look-ahead carry function. This device can be used with the MC10181/MC10581 4-unit ALU directly, or with the MC10180/MC10580 dual arithmetic unit in any computer, instrumentation or digital communication application requiring high speed arithmetic operation on long words.

When used with the MC10181/MC10581, the MC10179/MC10579 performs a second order or higher look-ahead. Figure 2 shows a 16-bit look-ahead carry arithmetic unit. Second order carry is valuable for longer binary words. As an example, addition of two 32-bit words is improved from 30 nanoseconds with ripple-carry techniques, to 18 nanoseconds with carry look-ahead techniques. A block diagram of a 32-bit ALU is shown in Figure 1. The MC10179/MC10579 may also be used in many other applications. It can, for example, reduce system package count when used to generate functions of several variables.

Numbers at ends of terminals denote pin numbers for L and P packages.
 Numbers in parenthesis denote pin numbers for F package.

Characteristic	Symbol	-55°C		-30°C		+25°C		+85°C		+125°C		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Power Supply Drain Current	I_E	—	79	—	79	—	72	—	79	—	79	mAdc
Input Current	I_{inH}	—	380	—	360	—	225	—	225	—	225	μAdc
Pins 5,9		—	460	—	430	—	270	—	270	—	270	
Pins 4,7,11		—	600	—	565	—	355	—	355	—	355	
Pin 14		—	670	—	630	—	395	—	395	—	395	
Pin 12		—	750	—	700	—	440	—	440	—	440	
Switching Times												ns
Propagation Delay	t_{pd}											
G or C_n to Carry; G or P to G_G		1.0	5.9	1.0	5.8	1.0	5.5	1.0	6.1	1.0	6.4	
P to P_G		1.0	3.9	1.0	3.7	1.0	3.5	1.0	3.9	1.0	4.1	
Rise Time, Fall Time (20% to 80%)	t^+, t^-	1.0	3.9	1.1	3.7	1.1	3.5	1.1	3.9	1.0	4.1	ns

-55°C and +125°C test values apply to MC105xx devices only.

FIGURE 1 – 32-BIT ALU WITH CARRY LOOK-AHEAD

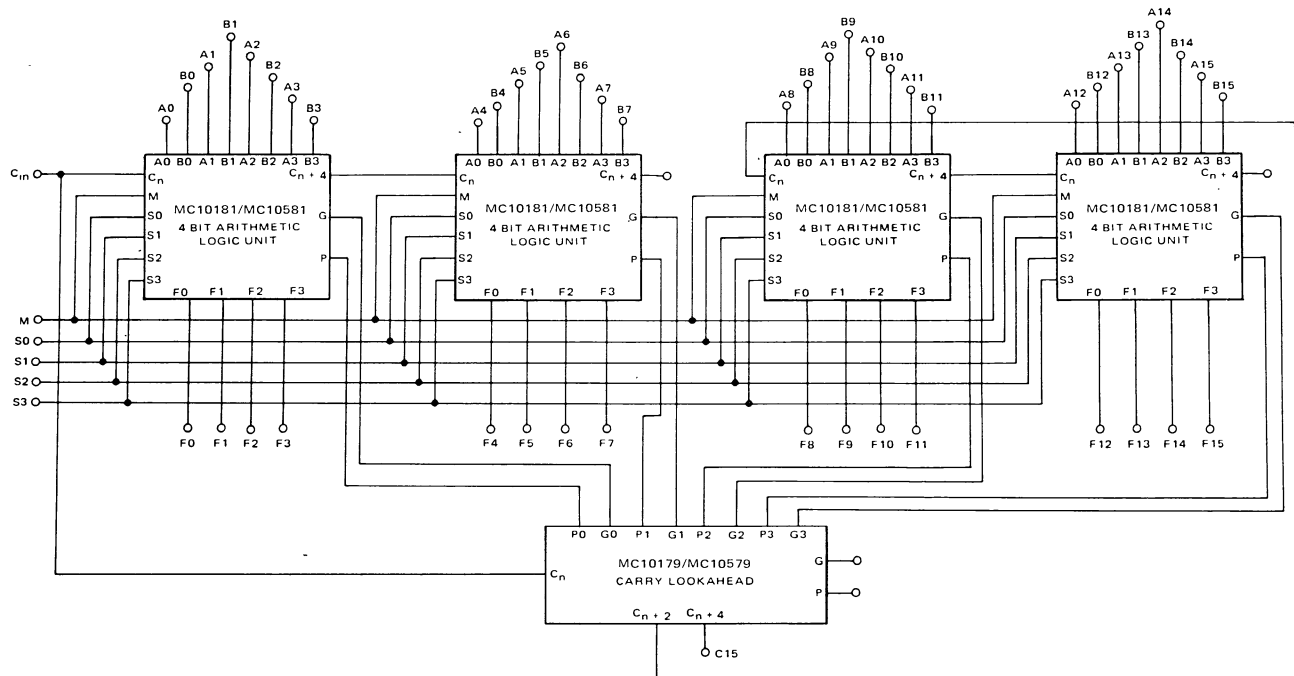
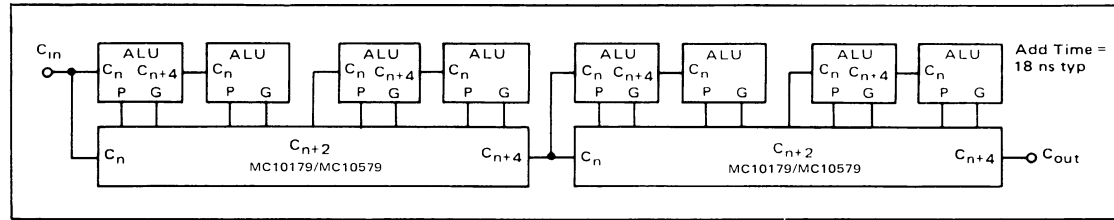


FIGURE 2 – 16-BIT FULL LOOK-AHEAD CARRY ARITHMETIC LOGIC UNIT

